

WATERSHED MANAGEMENT

Last Five Previous Year Question

- 1. Haryali is a scheme of government of India. This scheme was started with which of the following objective.**
 - a. For improving soil health
 - b. For integrated watershed development of a region with the inclusion of gram panchayat.
 - c. For integrated nutrient management
 - d. None of the above

- 2. Micro watershed is the watershed which is having _____ ha area under it.**
 - a. Less than 500 ha
 - b. More than 500 ha
 - c. Between 100 to 1000 ha area
 - d. Less than 100 ha

- 3. Mega/Macro watershed covers _____ ha region.**
 - a. More than 10,000ha
 - b. More than 50,0000 ha
 - c. Less than 10,000 ha
 - d. None of the above

- 4. Neeru- meeru scheme is implemented by government of india with the following objective.**
 - a. Nutrient conservation
 - b. Water conservation
 - c. Soil conservation
 - d. All of the above

Answers

1. B 2.C 3.B 4.B

WATERSHED MANAGEMENT

Watershed is **defined as a geohydrological unit draining to a common point by a system of drains**. All lands on earth are part of one watershed or other. Watershed is thus the land and water area, which contributes runoff to a common point.

A watershed is an area of land and water bounded by a drainage divide within which the surface runoff collects and flows out of the watershed through a single outlet into a larger river (or) lake.

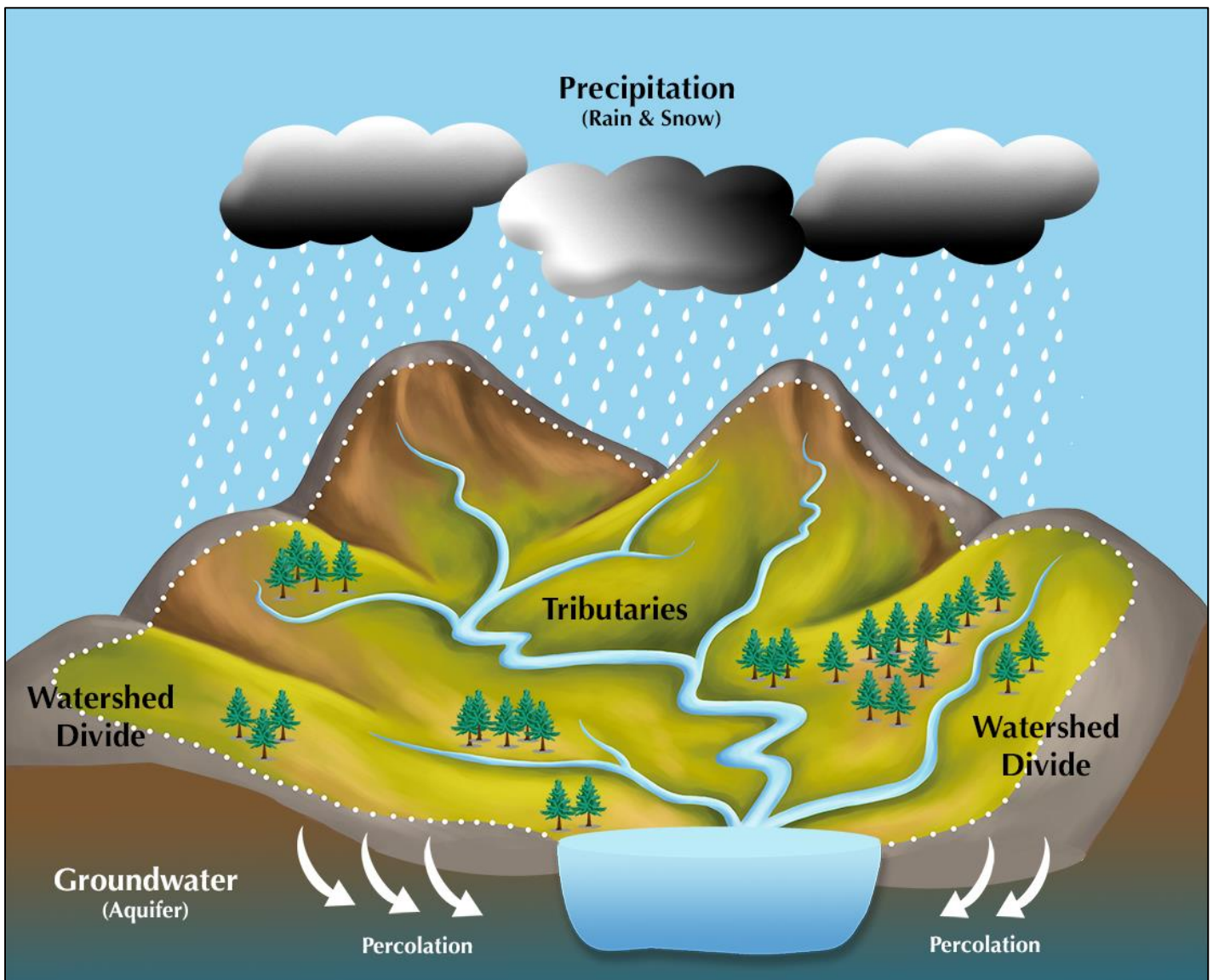


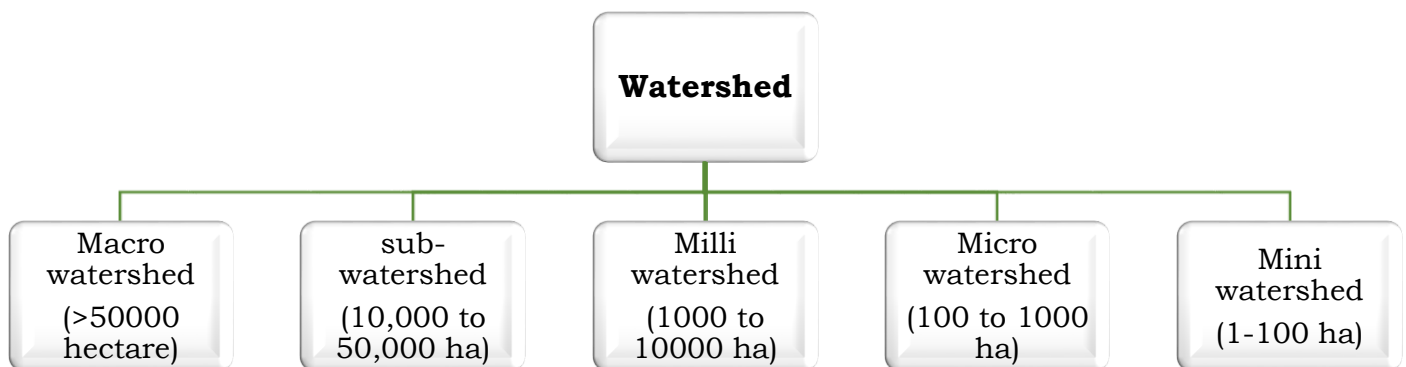
Figure 1 – Watershed

Types Of Watersheds In India

- A national system of Watershed Delineation and Codification has been developed by the Department of Agriculture's All India Soil and Land Use Survey Organization.
- The watershed **has been split into four categories**, depending on its size namely: Basins, Catchments, Sub-catchments, and Watersheds.
- There **are 3200 watersheds, 112 catchments**, 35 rivers, and 500 sub-catchments in India.

Watersheds is classified depending upon the size, drainage, shape and land use pattern.

1. **Macro watershed** = (> 50,000 Hect)
2. **Sub-watershed** = (10,000 to 50,000 Hect)
3. **Milli-watershed** = (1000 to 10000 Hect)
4. **Micro watershed** = (100 to 1000 Hect)
5. **Mini watershed** = (1-100 Hect)



Watershed Management - Principles

The core principles of watershed management are as follows:

- **Using the land in accordance with its potential** in light of land use classification.
- **Conserving as much rainfall** as possible in situ, or in the spot where it falls.
- **Removing extra water at a safe speed** and redirecting it to a storage area (like a tank or pond on a farm) for later use.

- **Preventing the creation of gullies** and building check dams in appropriate locations to manage soil erosion, collect rainwater, and rehydrate the groundwater.
- Determining **the best cropping system** for the watershed.
- **Achieving the most productivity** possible per unit of space, time, and water.

Objectives Of Watershed Management

The different objectives of watershed management programmes are:

- To **control damaging runoff** and degradation and thereby conservation of soil and water.
- To manage and **utilize the runoff water** for useful purpose.
- To protect, **conserve and improve the land of watershed** for more efficient and sustained production.
- To protect and **enhance the water resource originating** in the watershed.
- **To check soil erosion and** to reduce the effect of sediment yield on the watershed
- To **rehabilitate the deteriorating lands**.
- To moderate the **floods peaks at downstream areas**.
- To increase **infiltration of rainwater**.
- To improve and increase **the production of timbers, fodder and wild life resource**.
- To enhance the **ground water recharge, wherever applicable**

General Practices adopted in watershed management

1. **Cropping System**-crop rotation with the inclusion of pulse crop in between, high density crop, integrated nutrient management, integrated weed management and pest management, zero tillage etc should be adopted to conserve the soil moisture.
2. **Prevention Of Soil Erosion** by water by adopting various mechanical and agronomical measures like contour bunding, bench terracing, gabian structure etc. they can reduce the soil erosion as well as rainwater harvesting.
3. **Rain Water Harvesting** to reduce the wastage of rain water as a runoff and these water conservation practices are adopted in dryland region so that water can be used during season when no rainfall is there ex khadin in Rajasthan
4. **Integration of rural people, garm panchayat** and all the stakeholders so that proper implementation of policies should be there.

- 5. Integration of various agricultural activities** should be there like fisheries, animal husbandry etc so that water resources can be used efficiently and for the socio-economic development of that region.
- 6. Land rehabilitation** by agroforestry, afforestation and other conservation measures to be adopted to rehabilitate the degraded land.

Watershed Management - Significance

- **Controls Pollution**: Runoff from snowmelt or rains can add a large quantity of pollution to a lake or river. Watershed management thus aids in reducing contamination of the watershed's natural resources, including its water.
- **Identifies and Regulates Ecologically Hazardous Activities**: Every activity that takes place inside a watershed has some impact on the water's natural resources.
- Such activities are thoroughly identified in watershed management planning, and recommendations are made on how to effectively handle them in order to lessen their negative effects.
- **Enhances Partnership Among the Stakeholders**: It enhances the partnership among all watershed stakeholders as a result of watershed management planning which is crucial for the effective management of land and water resources.
- **Inclusive Growth**: Management of watersheds is essential for inclusive and sustainable growth. For instance, watershed management has demonstrated the **capacity to boost water availability, diversify crops and farming systems**, and support rural families by increasing agricultural production and providing varied revenue sources in drought-prone rainfed areas.
- **Improve Water Quality**: Water collected **directly from rain** is contaminated and useless; thus, Watershed Management can help improve water quality, which can then be used in households, agriculture, and industry.
- **Agriculture Productivity**: Watershed **management increases agricultural productivity**; it gives farmers more options by offering a variety of cropping systems, which can ultimately increase rural farmers' income and make the land suitable for further agriculture. **It also helps to improve agriculture in drought-prone areas** where rain is less likely; in such cases, land can make use of rain from another region and thus support farmers in their livelihood, making their lives easier.

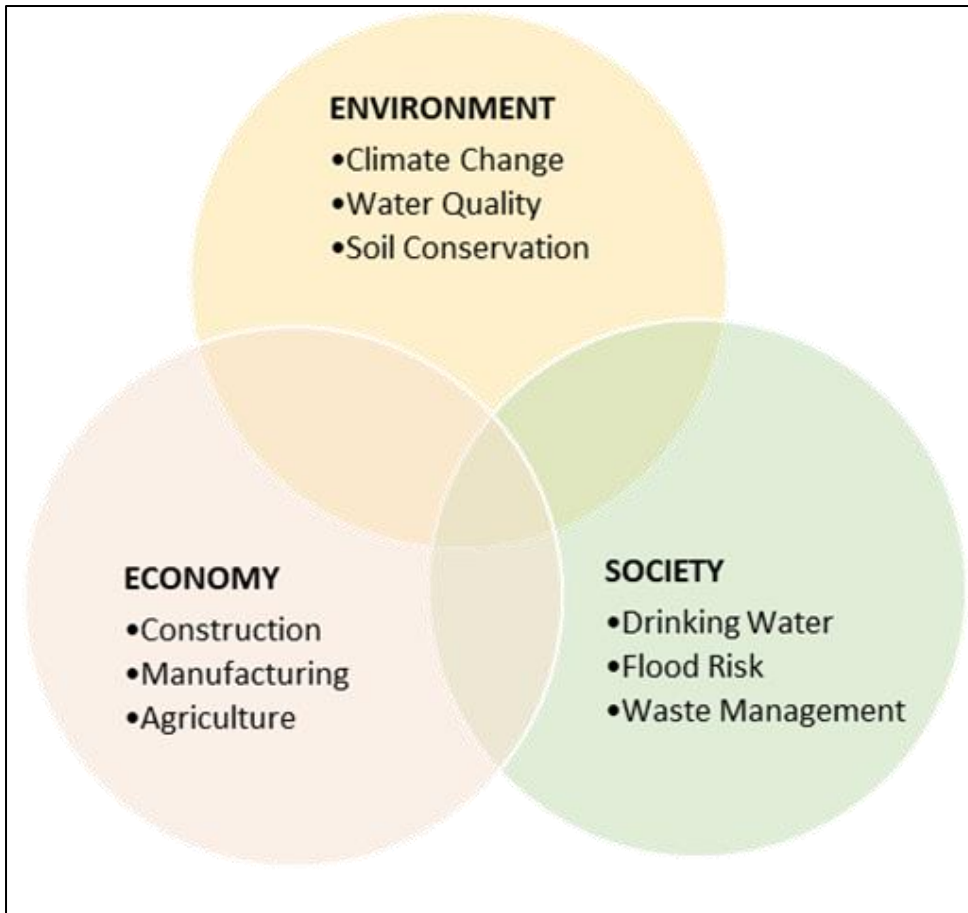
- **Economic Growth**: It is the most **promising way to use waste forest land to generate livelihood, income, and jobs for many people**. This method allows us to use land for a variety of purposes, which benefits the country's economy.

Watershed Management - Challenges

- **Project Related Issues**: The underperformance of watershed management programs **has been attributed to a number of factors, including outdated approaches, poor project design, insufficient and/or unsustainable financial resources**, very short time frames for project interventions, and a lack of adequate knowledge of the connections between upland and lowland areas.
- **Lack of Legal Support**: Although many nations have **broad environmental regulations in place**, the creation of watershed management policies rarely receives any attention. **The main obstacles to implementing sustainable watershed management programs** are acknowledged to be the absence or inadequacy of national policies, strategies, and action plans.
- **Weak Institutional Base**: Once these programs are over, it has been observed that **watershed-based institutions are on the verge of collapsing** since the inputs they got are frequently insufficient to support the institutional framework. Self-Help Groups are also not effectively incorporated into the watershed programs.
- **The sustainability of natural resources**, communities' capacity for diversification, and their ability to receive assistance from various programs are all influenced by the institutional base.
- **Initiatives Excluding Reserved Forestlands**: The establishment of entitlements over the forest **produce and the inclusion of reserve forestlands in watershed development plans are not yet included in watershed development programs**.

Integrated Watershed Management Programme
(IWMP)

The Integrated Watershed Management Programme (IWMP), which was launched in 2009–10 by the Ministry of Rural Development, aims to encompass 55 million hectares of rain-fed land by 2027.



**Figure 2 –
Components
of integrated
watershed
management**

The Department of Land Resources has been implementing it since 2009-10, after integrating three area development programmes, namely:

- Desert Development Programme (DDP)
- Drought Prone Areas Programme (DPAP)
- Integrated Wastelands Development Programme (IWDP)

After China's, the IWMP of India is the largest watershed program in the world.

The main goals of the IWMP are **to harness, conserve, and develop degraded natural resources such as soil, vegetative cover, and water**; to prevent soil run-off; to collect rainwater and recharge the groundwater table; to increase crop productivity; to introduce multi-cropping and diverse agro-based activities; to promote sustainable livelihoods, and to increase household incomes.

The initiative is being carried out in every state in the nation, and it is being funded 90:10 by the central and state governments.

Gram Panchayat has been effectively involved in the IWMP to perform important functions at the Village / Watershed level to maintain transparency and people's participation.

Watershed Management - Related Initiatives

- **Haryali Watershed Development Project:** The **Central Government is funding the Haryali watershed development project**, which intends to make it possible for the rural population to save water for drinking, irrigation, fisheries, and afforestation. With the assistance of the **public, Gram Panchayats** are carrying out this project.
- **Neeru-Meeru (Water and You) Program:** With the help of the public, the Neeru-Meeru (Water and You) initiative in Andhra Pradesh and Arvary Pani Sansad in Alwar, Rajasthan, have begun building various water-harvesting structures like percolation tanks, dug-out ponds (Jihad), check dams, etc.
- **Tamil Nadu's Mandate for water Harvesting Structures:** In Tamil Nadu, water harvesting systems for homes are required. Without building structures for water harvesting, no building can be built.